

Using Double Machine Learning to Estimate the Causal Effect of Pre-Primary Education on Student Achievement

Evidence from PISA 2022 in Germany

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Github Repository:

<https://github.com/julianlohmann/CausalMLproject>

Analytical Framework: Causal Effect of Pre-primary School Education

Data & Variables

General data description

Data: German subsample of PISA 2022 representative sample of all 15-year-old students

Sample: Complete-case sample ($N = 1,731$)

$N_{\text{no pre-schooling}} = 803$; $N_{\text{pre-schooling}} = 928$.

Variable overview

Treatment (T):

Pre-primary education: binary

Outcome (Y):

Reading ability: continuous

Covariates (X):

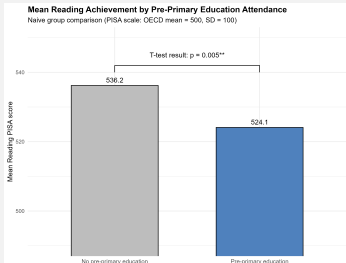
Student: gender, age, migration

Parents: SES, education, attitudes, migration

Collider (Z):

A lot of potential colliders and mediators in the dataset - but hopefully all were excluded from the covariate list.

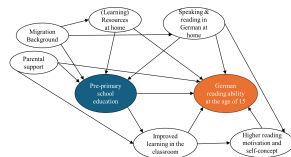
T-Test Baseline



Naïve conclusion

Protect your children from pre-primary school education!

Structural Model



First Naïve Findings

T-test implies that pre-primary education reduces reading ability by around 12 PISA points on average.

This result may be confounded, motivating a reanalysis with Double ML